

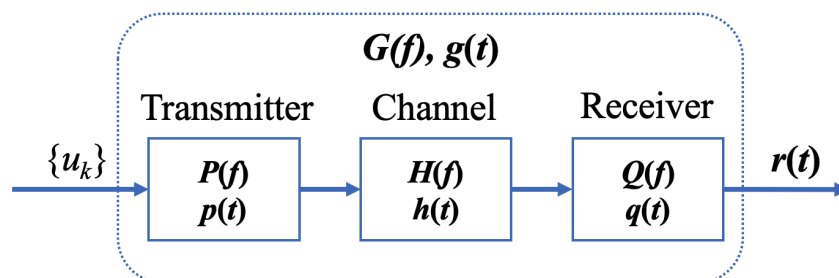
Communications and Networks (Spring 2022)

Problem Set 3

Issued: Mar. 28 Due: Apr. 10

Readings: Proakis - Digital Communications 3.2, 9.2.1
Haykin - Communication System 5.9-5.10, 8.2

1. A speech signal is sampled at a rate of 8 kHz, and then encoded using 8 bits per sample. The binary stream is then transmitted through an AWGN baseband channel via M-level PAM (Pulse Amplitude Modulation). The ISI (Inter-Symbol Interference) should be eliminated.
 - a) Determine the minimum bandwidth required for transmission when $M = 4$.
 - b) Determine the minimum bandwidth required for transmission when $M = 16$. What is the price to pay for saving the bandwidth?
2. For a 4-level PAM baseband transmission system, the transmitted signals consists of pulses having a raised cosine spectrum with a roll-off factor 0.5 and its spectrum extends to 7.5kHz.
 - a) What is the maximum data rate with zero-ISI requirement? What is the corresponding maximum bandwidth efficiency?
 - b) Will there be any ISI when the symbol period is 0.2ms? What about 0.15ms?
Explain in the perspectives of both the time domain and the frequency domain.
3. Consider a PAM baseband transmission system. The baseband waveform is $p(t)$ with the symbol interval T_s , the channel is characterized by a filter $h(t)$, and the filter at the receiver is $q(t)$. As a result, the received waveform is given by $r(t) = \sum_k u_k g(t - kT)$, where $g(t) = p(t) * h(t) * q(t)$ and u_k is the sequence of symbols.



- a) What property must $g(t)$ have, so that $r(kT) = u_k$ for any k ?
- b) Recall the Fourier transform of $\text{sinc}(x)$ is

$$\mathcal{F}\{\text{sinc}(x)\} = \text{rect}(f) = \begin{cases} 1, & f \in [-0.5, 0.5] \\ 0, & \text{else} \end{cases}.$$

Let the transmitter waveform be $p(t) = \text{sinc}(1000t)\text{sinc}(2000t)$. Calculate the frequency domain representation $P(f)$ of the waveform.

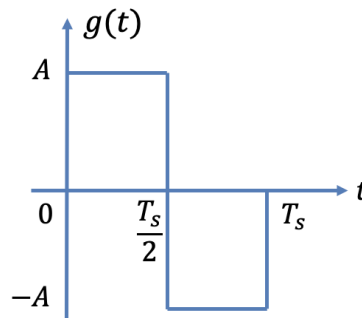
- c) Assume the symbol time is $T_s = 0.5\text{ms}$, and the frequency response of the channel is specified by

$$H(f) = \begin{cases} 1, & |f| \in [0, 0.75\text{kHz}] \\ 0, & |f| \in (0.75\text{kHz}, 1\text{kHz}] \\ 1, & |f| \in (1\text{kHz}, 1.25\text{kHz}] \\ 0, & |f| \in (1.25\text{kHz}, \infty] \end{cases}.$$

Can you design a receiver filter $Q(f)$ so that there is no ISI?

(Hint: The spectrum $G(f)$ of $g(t)$ should satisfy the condition that $G(f) + G(1/T_s - f)$ is a constant, according to the Nyquist criterion)

- d) If $H(f)$ is non-zero only at $[0, 0.75\text{kHz}]$, is it possible to design a no-ISI receiver and why?
4. In a baseband transmission system, the output waveform of the transmitter is given below. The channel is additive white Gaussian noise (AWGN) channel with noise PSD $n_0/2$.



- a) Design a matched filter for the optimal receiver and draw the impulse response of the filter.
- b) Draw the output of the matched filter.
- c) Compute the optimal SNR at the sample output after the matched filter.

5. In this problem, we derive the matched filter in the time domain.

Suppose a symbol a_i is modulated on a baseband pulse waveform $g(t)$. The receiver filter is $h(t)$ and is sampled at $t = T_s$. Consider an additive white Gaussian noise $n(t)$ with PSD (Power Spectral Density) $n_0/2$. The sampled signal is:

$$Y = S + N,$$

where $S = a_i \int_{-\infty}^{\infty} g(\tau)h(T_s - \tau) d\tau$, $N = \int_{-\infty}^{\infty} n(\tau)h(T_s - \tau) d\tau$.

a) Prove the expression of the Signal-to-Noise Ratio (SNR) at the output sample point is

$$\text{SNR} = \frac{S^2}{\mathbb{E}[N^2]} = \frac{2a_i^2}{n_0} \frac{\left[\int_{-\infty}^{\infty} g(\tau)h(T_s - \tau) d\tau \right]^2}{\int_{-\infty}^{\infty} h^2(\tau) d\tau}$$

b) Find the maximum SNR and the corresponding $h(t)$.

(Hint: use the Cauchy-Schwarz inequality: $(\int f(x)g(x) dx)^2 \leq \int f^2(x) dx \int g^2(x) dx$, with equality if and only if $g(x) = kf(x)$)